Sales Data Analysis using Excel, Python & SQL

Ayesha Wasim
B.Tech CSE (6th Semester)
Enrollment No: 2023-310-065
Jamia Hamdard

Introduction

- Data analysis is important for understanding business performance.
- This project focuses on analyzing sales data to identify trends, patterns, and insights using Excel, Python, and SQL.

Objective

To analyze sales data

To identify best and worst performing products

To understand city-wise sales trends

To use Excel, Python, and SQL for data analysis

Problem Statement

Businesses generate large amounts of sales data but often fail to analyze it effectively.

This leads to poor decision-making.

This project focuses on analyzing data using Excel, Python, and SQL to extract useful insights.

TOOLS & TECHNOLOGIES

Microsoft Excel

Python

SQL

Charts & Visualization

Dataset Description

The dataset includes:

- Product Name
- Price
- Quantity
- City
- Total Sales

EXCEL ANALYSIS

- Used SUM, AVERAGE, MAX, MIN Functions
- Created Pivot Tables for data summarization
- Generated (Bar & Pie) charts for visualization
- Performed city-wise and product-wise analysis

Excel Analysis & Visualization

A	B	C	D	E	F	G	H	I	J
Order_ID	Customer	Product	Category	Price	Quantity	City	Payment	Order_Dat	Total
101	Saara	Shoes	Fashion	1200	2	Delhi	UPI	1/4/2026	2400
102	Rehmaan	Watch	Accessori	1800	1	Mumbai	Card	2/4/2026	1800
103	Urooj	Bag	Fashion	1500	3	Jaipur	Cash	3/4/2026	4500
104	Kamran	Shirt	Clothing	900	2	Rajisthan	UPI	4/4/2026	1800
105	Abdullah	Laptop	Electronic	700	4	Hyderaba	Card	5/4/2026	2800
106	Ibrahim	Bottle	Home	200	5	Noida	UPI	5/4/2026	1000
107	Mehak	Headphor	Electronic	2200	1	Gurgaon	Card	7/4/2026	2200
108	Umar	Jeans	Clothing	1400	2	Banglore	Cash	8/4/2026	2800
109	Umair	Shoes	Fashion	1200	1	Karnataka	Cash	9/4/2026	1200
110	Zoha	Watch	Accessori	1800	4	Noida	UPI	10/4/2026	7200
								Total Sales	27700
								Average Sa	2770
								Max Sales	7200
								Min Sales	1000
								Delhi Orde	1
								Delhi Total	2400
								Delhi Aver	2400

SQL Analysis

- Used SQL for analyzing sales data
- Applied aggregate functions like SUM, COUNT, and AVG
- Identified highest sales and category-wise averages
- Performed city-wise and product count analysis
- Extracted meaningful insights from structured data

City-wise sales

Input

Run SQL

```sql
SELECT City, SUM(Total) AS Total_sales
FROM sales_data
GROUP BY City;
```

Output

City	Total_sales
Banglore	2800
Delhi	2400
Gurgaon	2200
Hyderabad	2800
Jaipur	4500
Karnataka	1200
Mumbai	1800
Noida	8200
Rajasthan	1800

Available Tables

Customers

customer_id	first_name	last_name	age	country
empty				

Orders

order_id	item	amount	customer_id
empty			

Sales_data

Order_ID	Customer_Name	Product	Category	Price	Quantity	City
101	Saara	Shoes	Fashion	1200	2	Delhi
102	Rehmaan	Watch	Accessories	1800	1	Mumb
103	Urooj	Bag	Fashion	1500	3	Jaipu
104	Kamran	Shirt	Clothing	900	2	Rajas
105	Abdullah	Laptop	Electronic	700	4	Hyde
106	Ibrahim	Bottle	Home	200	5	Noida
107	Mehak	Headphone	Electronic	2200	1	Gurg
108	Umar	Jeans	Clothing	1400	2	Bang
109	Umair	Shoes	Fashion	1200	1	Karna
110	Zoha	Watch	Accessories	1800	4	Noida

Shippings

shipping_id	status	customer
empty		

Python Implementation

Developed interactive program using user input
Applied variables and input/output concepts
Generated dynamic output based on user data

Calculator using user input

Even/Odd number checker

Simple data operations

Variables and Data Types

Conditional Statements (if-else)

Input/Output Handling

main.py

Share Run

```python
name = input("Enter your name: ")
age = input("Enter your age: ")
study = input("Enter your study: ")
job = input("Enter your job: ")

print("Hello " + name + ", You are " + age + ", You study " + study + ", Your job is " + job)
```

Output Clear

```
Enter your name: Ayesha Wasim
Enter your age: 21
Enter your study: Btech CSE
Enter your job: MIS Executive
Hello Ayesha Wasim, You are 21, You study Btech CSE, Your job is MIS Executive

=== Code Execution Successful ===
```

ER Diagram

The system consists of three entities:

Product

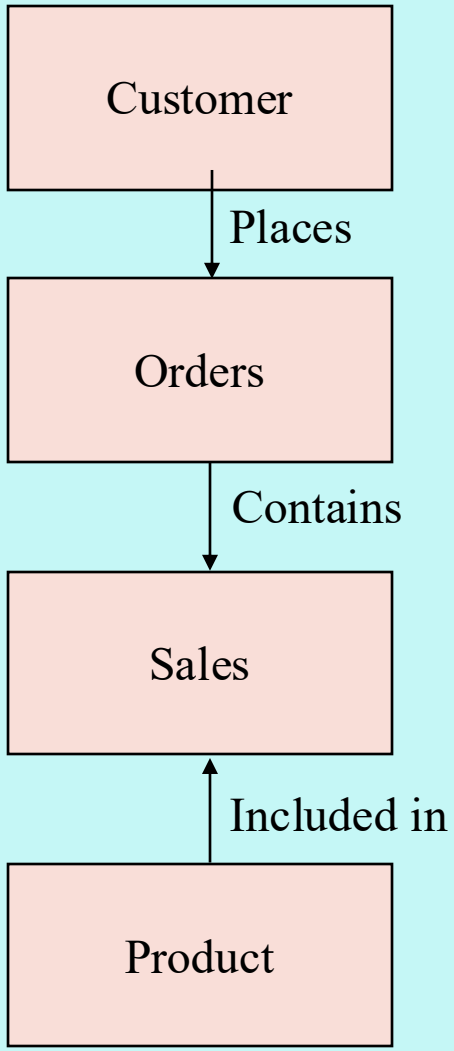
Sales

City

Sales acts as a connecting entity between Product and City.

Entity Relationship Diagram(ER Diagram)

This ER diagram represents the relationship between customers, orders, products, and sales data in the system.



RESULTS / INSIGHTS

- Some products have higher sales than others
- Certain cities generate more revenue
- Data visualization helped in better understanding
- Patterns in sales were identified

Conclusion

This project shows how Excel, Python, and SQL can be used together for data analysis.

It helps in understanding business data and improving decision-making.

THANK YOU